

Lecture Notes: Counting k -Dimensional Subspaces of an n -Dimensional Vector Space

Contents

1	The Main Question	2
2	Ordinary Binomial Coefficients as Motivation	2
3	Statement of the Formula over a Finite Field	2
4	Proof by Counting Ordered Bases	3
4.1	Counting Ordered Linearly Independent k -Tuples in $\mathbf{F}_q^n$	3
4.2	Counting Ordered Bases of a Fixed k -Dimensional Subspace	3
4.3	Dividing Out the Overcount	3
5	Why the Denominator Appears	4
6	Examples	4
7	Special Cases	5
8	Symmetry	5
9	A Recursive Formula	5
10	Polynomial Nature	6
11	The $q \rightarrow 1$ Philosophy	6
12	What Happens over Infinite Fields?	7
13	The Grassmannian Viewpoint	7
14	A Matrix-Theoretic Interpretation	7
15	Related Formula: The Size of $\text{GL}_n(\mathbf{F}_q)$	8
16	Conceptual Summary	8
17	Final Theorem	8
18	Exercises	8
19	Selected Answers	9

1 The Main Question

Let V be an n -dimensional vector space. We want to answer the following question:

How many distinct k -dimensional subspaces does V have?

The answer depends on the field over which V is defined.

- If the field is finite, say $\mathbf{F}_q$ with q elements, then the answer is finite and is given by the *Gaussian binomial coefficient*.
- If the field is infinite, such as $\mathbf{R}$, $\mathbf{C}$, or $\mathbf{Q}$, then there are infinitely many k -dimensional subspaces whenever $0 < k < n$.

These notes explain the finite-field formula, prove it carefully, and place it in a broader geometric and combinatorial context.

2 Ordinary Binomial Coefficients as Motivation

Recall that the number of k -element subsets of an n -element set is

$$\binom{n}{k}.$$

The Gaussian binomial coefficient is the vector-space analogue of this number.

Set Theory	Linear Algebra over $\mathbf{F}_q$
k -element subsets of an n -element set $\binom{n}{k}$	k -dimensional subspaces of $\mathbf{F}_q^n$ $\begin{bmatrix} n \\ k \end{bmatrix}_q$

So the Gaussian binomial coefficient is often called a q -analogue of the ordinary binomial coefficient.

3 Statement of the Formula over a Finite Field

Assume from now on that

$$V \cong \mathbf{F}_q^n,$$

where $\mathbf{F}_q$ is a finite field with q elements.

Definition 3.1. The *Gaussian binomial coefficient* (or *q -binomial coefficient*) is defined by

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^n - 1)(q^n - q)(q^n - q^2) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q)(q^k - q^2) \cdots (q^k - q^{k-1})}$$

for $0 \leq k \leq n$.

An equivalent product form is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{i=0}^{k-1} \frac{q^n - q^i}{q^k - q^i}.$$

Another useful form is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^n - 1)(q^{n-1} - 1) \cdots (q^{n-k+1} - 1)}{(q^k - 1)(q^{k-1} - 1) \cdots (q - 1)}.$$

Theorem 3.2. Let V be an n -dimensional vector space over $\mathbf{F}_q$. Then the number of distinct k -dimensional subspaces of V is

$$\boxed{\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{i=0}^{k-1} \frac{q^n - q^i}{q^k - q^i}.$$

4 Proof by Counting Ordered Bases

The proof is a clean double-counting argument.

4.1 Counting Ordered Linearly Independent k -Tuples in $\mathbf{F}_q^n$

We count the number of ordered tuples

$$(v_1, \dots, v_k) \in (\mathbf{F}_q^n)^k$$

such that $v_1, \dots, v_k$ are linearly independent.

- The first vector v_1 can be any nonzero vector, so there are $q^n - 1$ choices.
- The second vector v_2 must not lie in $\text{span}(v_1)$, and $\text{span}(v_1)$ has q elements, so there are $q^n - q$ choices.
- The third vector v_3 must not lie in $\text{span}(v_1, v_2)$, which has q^2 elements, so there are $q^n - q^2$ choices.

Continuing in this way, we obtain:

Lemma 4.1. *The number of ordered linearly independent k -tuples in $\mathbf{F}_q^n$ is*

$$(q^n - 1)(q^n - q)(q^n - q^2) \cdots (q^n - q^{k-1}).$$

4.2 Counting Ordered Bases of a Fixed k -Dimensional Subspace

Now fix a k -dimensional subspace $W \subseteq \mathbf{F}_q^n$. Since $W \cong \mathbf{F}_q^k$, the number of ordered bases of W is the same as the number of ordered linearly independent k -tuples in $\mathbf{F}_q^k$.

Lemma 4.2. *A fixed k -dimensional subspace W has exactly*

$$(q^k - 1)(q^k - q)(q^k - q^2) \cdots (q^k - q^{k-1})$$

ordered bases.

4.3 Dividing Out the Overcount

Each ordered linearly independent k -tuple spans a unique k -dimensional subspace, and every k -dimensional subspace contributes exactly the number of ordered bases computed above.

Therefore:

Proof of the Theorem. The number of ordered linearly independent k -tuples in $\mathbf{F}_q^n$ is

$$(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1}).$$

Each k -dimensional subspace has exactly

$$(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})$$

ordered bases. Dividing, we obtain the number of distinct k -dimensional subspaces:

$$\frac{(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})} = \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

□

5 Why the Denominator Appears

A frequent source of confusion is the denominator

$$(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1}).$$

The reason is that many different ordered bases span the same subspace. So the numerator counts the same subspace many times, once for each ordered basis of that subspace.

Thus:

- numerator = all ordered bases of all k -dimensional subspaces,
- denominator = ordered bases of one fixed k -dimensional subspace,
- quotient = number of distinct k -dimensional subspaces.

6 Examples

Example 6.1 (Lines in $\mathbf{F}_q^3$). A 1-dimensional subspace is a line through the origin. The number of such lines is

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix}_q = \frac{q^3 - 1}{q - 1} = q^2 + q + 1.$$

Example 6.2 (Planes in $\mathbf{F}_q^3$). A 2-dimensional subspace of $\mathbf{F}_q^3$ is a plane through the origin. The number of such planes is

$$\begin{bmatrix} 3 \\ 2 \end{bmatrix}_q = \begin{bmatrix} 3 \\ 1 \end{bmatrix}_q = q^2 + q + 1.$$

Example 6.3 (Two-dimensional subspaces of $\mathbf{F}_2^4$). Take $q = 2$, $n = 4$, $k = 2$. Then

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q = \frac{(2^4 - 1)(2^4 - 2)}{(2^2 - 1)(2^2 - 2)} = \frac{15 \cdot 14}{3 \cdot 2} = 35.$$

So $\mathbf{F}_2^4$ has exactly 35 distinct 2-dimensional subspaces.

Example 6.4 (All lines in $\mathbf{F}_2^2$). The vector space $\mathbf{F}_2^2$ has vectors

$$(0, 0), \quad (1, 0), \quad (0, 1), \quad (1, 1).$$

Its nonzero vectors are

$$(1, 0), \quad (0, 1), \quad (1, 1).$$

Over $\mathbf{F}_2$, each 1-dimensional subspace has exactly one nonzero vector, so the distinct lines are

$$\text{span}(1, 0), \quad \text{span}(0, 1), \quad \text{span}(1, 1).$$

Thus there are 3 of them, in agreement with

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}_q = \frac{2^2 - 1}{2 - 1} = 3.$$

7 Special Cases

Proposition 7.1. *For all n and q , we have:*

$$\begin{aligned}\begin{bmatrix} n \\ 0 \end{bmatrix}_q &= 1, \\ \begin{bmatrix} n \\ 1 \end{bmatrix}_q &= \frac{q^n - 1}{q - 1} = 1 + q + q^2 + \cdots + q^{n-1}, \\ \begin{bmatrix} n \\ n-1 \end{bmatrix}_q &= \begin{bmatrix} n \\ 1 \end{bmatrix}_q, \\ \begin{bmatrix} n \\ n \end{bmatrix}_q &= 1.\end{aligned}$$

Proof. The cases $k = 0$ and $k = n$ correspond to the unique zero subspace and the whole space. The case $k = 1$ is immediate from the formula. The case $k = n - 1$ follows from symmetry, proved below. $\square$

8 Symmetry

Just as ordinary binomial coefficients satisfy

$$\binom{n}{k} = \binom{n}{n-k},$$

Gaussian binomial coefficients satisfy the analogous identity.

Proposition 8.1. *For $0 \leq k \leq n$,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q.$$

Proof. Using the form

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^n - 1)(q^{n-1} - 1) \cdots (q^{n-k+1} - 1)}{(q^k - 1)(q^{k-1} - 1) \cdots (q - 1)},$$

one checks directly that replacing k by $n - k$ produces the same value after cancellation. $\square$

Remark 8.2. This symmetry reflects a geometric duality: a k -dimensional subspace is naturally related to an $(n - k)$ -dimensional object in the dual vector space.

9 A Recursive Formula

Ordinary binomial coefficients satisfy Pascal's identity

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

Gaussian binomial coefficients satisfy a q -analogue.

Proposition 9.1. *For $1 \leq k \leq n - 1$,*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q.$$

Proof. Fix a hyperplane $H \subseteq \mathbf{F}_q^n$ of dimension $n - 1$. Let $W \subseteq \mathbf{F}_q^n$ be a k -dimensional subspace. There are two cases.

1. $W \subseteq H$. The number of such subspaces is $\begin{bmatrix} n-1 \\ k \end{bmatrix}_q$.
2. $W \not\subseteq H$. Then $W \cap H$ has dimension $k - 1$. For each $(k - 1)$ -dimensional subspace $U \subseteq H$, the number of k -dimensional subspaces W such that $W \cap H = U$ is q^{n-k} . Hence the number of subspaces in this case is

$$q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q.$$

Adding the two cases gives

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q.$$

□

10 Polynomial Nature

Although the formula for $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is written as a fraction, it actually simplifies to a polynomial in q with integer coefficients.

Example 10.1.

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix}_q = q^2 + q + 1.$$

Also,

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q = q^4 + q^3 + 2q^2 + q + 1.$$

This fact is highly important in algebraic combinatorics.

11 The $q \rightarrow 1$ Philosophy

The Gaussian binomial coefficient is called a q -analogue of $\binom{n}{k}$ because, after simplification as a polynomial in q , substituting $q = 1$ gives the ordinary binomial coefficient.

Example 11.1. Since

$$\begin{bmatrix} 3 \\ 1 \end{bmatrix}_q = q^2 + q + 1,$$

setting $q = 1$ gives

$$1 + 1 + 1 = 3 = \binom{3}{1}.$$

So one may think of

$$\begin{bmatrix} n \\ k \end{bmatrix}_q$$

as the vector-space version of

$$\binom{n}{k}.$$

12 What Happens over Infinite Fields?

Suppose now that the base field is infinite.

Proposition 12.1. *If V is an n -dimensional vector space over an infinite field and $0 < k < n$, then V has infinitely many distinct k -dimensional subspaces.*

Proof. It is enough to consider the case $k = 1$. In $\mathbf{R}^2$, each line through the origin is a 1-dimensional subspace, and such lines are parametrized by slope:

$$y = mx \quad (m \in \mathbf{R}),$$

together with the vertical line $x = 0$. Thus there are infinitely many 1-dimensional subspaces. The general case follows similarly. $\square$

Remark 12.2. The only finite cases are trivial:

- exactly one 0-dimensional subspace, namely $\{0\}$,
- exactly one n -dimensional subspace, namely V .

13 The Grassmannian Viewpoint

Definition 13.1. The set of all k -dimensional subspaces of an n -dimensional vector space is called the *Grassmannian*, denoted

$$\text{Gr}(k, n).$$

Over a finite field, the Grassmannian is a finite set, and

$$|\text{Gr}(k, n)(\mathbf{F}_q)| = \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

Over $\mathbf{R}$ or $\mathbf{C}$, the Grassmannian is not finite. Instead, it is a geometric object: a manifold and also an algebraic variety.

So the Gaussian binomial coefficient counts the $\mathbf{F}_q$ -rational points of the Grassmannian.

14 A Matrix-Theoretic Interpretation

There is another very useful proof using matrices.

A k -dimensional subspace of $\mathbf{F}_q^n$ can be represented as the row space of a full-rank $k \times n$ matrix.

- The number of full-rank $k \times n$ matrices is

$$(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1}).$$

- Two such matrices have the same row space if and only if one is obtained from the other by multiplication on the left by an invertible $k \times k$ matrix.
- The group $\text{GL}_k(\mathbf{F}_q)$ has size

$$|\text{GL}_k(\mathbf{F}_q)| = (q^k - 1)(q^k - q) \cdots (q^k - q^{k-1}).$$

Therefore,

$$\#\{k\text{-dimensional subspaces of } \mathbf{F}_q^n\} = \frac{\#\{\text{full-rank } k \times n \text{ matrices}\}}{|\text{GL}_k(\mathbf{F}_q)|} = \begin{bmatrix} n \\ k \end{bmatrix}_q.$$

15 Related Formula: The Size of $\text{GL}_n(\mathbf{F}_q)$

The same counting method gives the size of the general linear group.

Proposition 15.1.

$$|\text{GL}_n(\mathbf{F}_q)| = (q^n - 1)(q^n - q)(q^n - q^2) \cdots (q^n - q^{n-1}).$$

Proof. An invertible $n \times n$ matrix is exactly an ordered basis of $\mathbf{F}_q^n$. The first column can be any nonzero vector, the second must be outside the span of the first, and so on. $\square$

This is the full-basis version of the same principle used in the Gaussian binomial coefficient.

16 Conceptual Summary

The formula

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q^n - 1)(q^n - q) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q) \cdots (q^k - q^{k-1})}$$

comes from two ideas:

1. Count ordered linearly independent k -tuples in $\mathbf{F}_q^n$.
2. Divide by the number of ordered bases of a fixed k -dimensional subspace.

This is the exact finite-field analogue of counting k -element subsets of an n -element set.

17 Final Theorem

Theorem 17.1 (Final Form). *Let V be an n -dimensional vector space over $\mathbf{F}_q$. Then the number of distinct k -dimensional subspaces of V is*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{i=0}^{k-1} \frac{q^n - q^i}{q^k - q^i}.$$

If the base field is infinite, then for $0 < k < n$ the number of k -dimensional subspaces is infinite.

18 Exercises

Exercise 18.1. Compute $\begin{bmatrix} 5 \\ 1 \end{bmatrix}_q$ when $q = 2$.

Exercise 18.2. Compute $\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q$ when $q = 3$.

Exercise 18.3. Show directly that

$$\begin{bmatrix} n \\ 1 \end{bmatrix}_q = 1 + q + q^2 + \cdots + q^{n-1}.$$

Exercise 18.4. Prove that

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q.$$

Exercise 18.5. Verify the recursion

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q.$$

Exercise 18.6. Count the 2-dimensional subspaces of $\mathbf{F}_2^3$.

Exercise 18.7. Explain why there are infinitely many 1-dimensional subspaces of $\mathbf{R}^2$.

19 Selected Answers

- For $q = 2$,

$$\begin{bmatrix} 5 \\ 1 \end{bmatrix}_q = \frac{2^5 - 1}{2 - 1} = 31.$$

- For $q = 3$,

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix}_q = \frac{(3^4 - 1)(3^4 - 3)}{(3^2 - 1)(3^2 - 3)} = \frac{80 \cdot 78}{8 \cdot 6} = 130.$$

-

$$\begin{pmatrix} 3 \\ 2 \end{pmatrix}_{q=2} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}_{q=2} = \frac{2^3 - 1}{2 - 1} = 7.$$

One-Sentence Takeaway

Over a finite field $\mathbf{F}_q$, the number of k -dimensional subspaces of $\mathbf{F}_q^n$ is the Gaussian binomial coefficient $\begin{bmatrix} n \\ k \end{bmatrix}_q$; over an infinite field, the number is usually infinite.